181127: quartz sandstone, Durba Hills

Location and sampling

GUNANYA (SG 51-14), DURBA (3350)
MGA Zone 51, 447465E 7377349N

Sampled on 25 September 2004

The sample was taken from the foot of the giant foreset exposed at the northern tip of Durba Hills, with preferential sampling of coarser grained sandstone to increase the proportion of possible heavy minerals. The locality is about 100 m southeast of the Canning Stock Route, and about 5 km northwest of the site of GSWA 181126 (Wingate and Bodorkos, 2007). Durba Hills is about 390 km (direct) north-northeast of Wiluna, and about 800 km (direct) southwest of Halls Creek.

Tectonic unit/relations

The unit sampled is a fine-grained, red and cream-coloured lithic arenite assigned to the McFadden Formation of the northwest Officer Basin. The McFadden Formation is dominantly an eolian deposit, with lesser fluvial facies, and is correlated (based on lithology and depositional facies) with the Lungkarta Formation in the subsurface of the Officer Basin to the east and southeast (Grey et al., 2005; Hocking et al., 2005). The McFadden Formation unconformably overlies the Late Neoproterozoic glacigene Boondawari Formation (Grey et al., 2005), and is disconformably overlain by the Durba Sandstone (GSWA 181126; Wingate and Bodorkos, 2007). A minimum age for the correlative Lungkarta Formation is provided by the overlying c. 510 Ma Table Hill Volcanics (Grey et al., 2005).

In GSWA drillhole Lancer 1, about 200 km to the east, there is no marked unconformity between sandstones assigned to the Lungkarta and Durba Sandstones, suggesting their ages (and therefore the ages of the McFadden and Durba Sandstones) may be similar (Haines et al., 2004; Grey et al., 2005).

Petrographic description

This is a coarse-grained sandstone that contains millimetre-scale bedding laminations, with maximum grain sizes varying from 0.25 mm to 1 mm between layers. Well-rounded single-crystal quartz grains are the main component, but there are also disseminated grains of microcline, clasts of polycrystalline quartz, and rarer fine-grained lithic fragments comprising plagioclase, quartz, clay minerals, opaque oxide minerals, leucoxene, and apatite in various proportions. Heavy minerals include opaque oxide minerals, tourmaline, and rare zircon. Interstitial clay minerals and hematite are abundant, with voids in some areas, and hematite-poor clay minerals in others.

The lithic fragments appear to be clasts derived from mafic and felsic volcanic rocks, or possibly granophyric rocks. Some of the polycrystalline quartz clasts are granular and/or microspherulitic, possibly indicating a low-temperature hydrothermal origin.

Zircon morphology

The zircons isolated from this sample are anhedral to subhedral, and moderately rounded to subspherical. Most are clear and colourless, although some are pale brown in colour. They are up to 500 μm long, with aspect ratios up to 4:1. The external surfaces of most zircons are pitted, consistent with sedimentary transport. Concentric growth zoning is common, and typically is truncated at (abraded) grain boundaries. Many zircons show complex internal structure, and a cathodoluminescence image of representative zircons is shown in Figure 1.

Analytical details

This sample was analysed on 21–22 October 2005 (using SHRIMP-A) and 24–25 October 2005 (using SHRIMP-B). Analyses 1.1 to 11.2 (spot numbers 1–21 inclusive) were obtained during the first session, together with 11 analyses of the CZ3 standard that indicated an external spot-to-spot (reproducibility) uncertainty of 1.71% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.56% (1σ). Analyses 20.1 to 55.1 (spot numbers 22–57 inclusive) were obtained during the second session, together with 18 analyses of the CZ3 standard that indicated an external spot-to-spot (reproducibility) uncertainty of 1.37% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.37% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and ages quoted in Table 1.

Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined

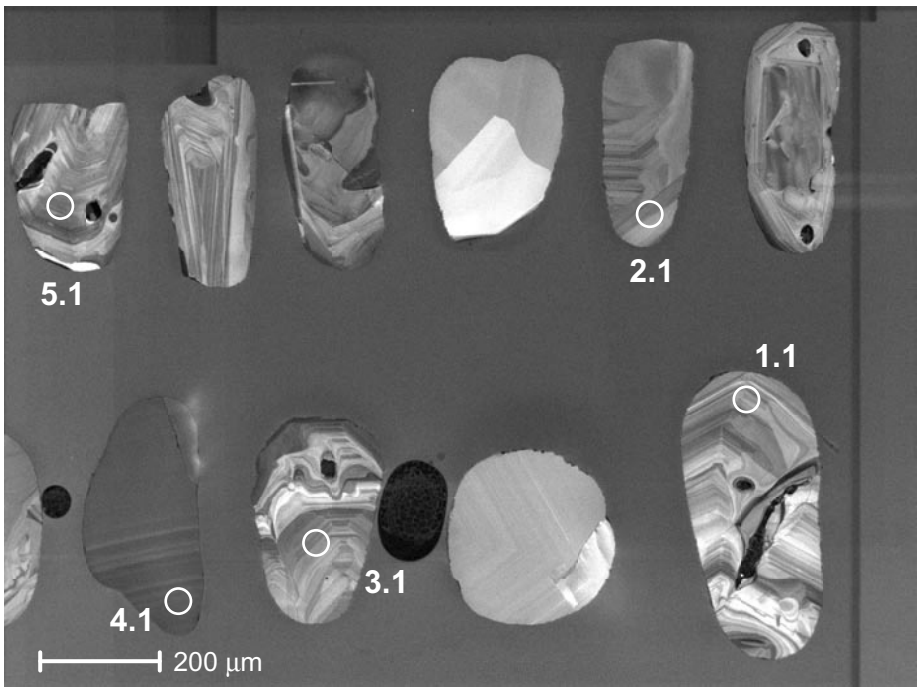

Figure 1. Cathodoluminescence image of representative zircons from sample 181127: quartz sandstone, Durba Hills. Numbered circles represent approximate positions of analysis sites

by the method of Stacey and Kramers (1975). However, the data were dispersed in $^{207}Pb^*/^{206}Pb^*$ when corrected for common Pb using measured ^{204}Pb (Fig. 2), and a negative correlation exists between the magnitude ($f204$) of the common-Pb correction and the $^{207}Pb^*/^{206}Pb^*$ dates (corrected for common Pb) in each of the analytical sessions (Fig. 3). This indicates that the common-Pb correction based on measured ^{204}Pb is inaccurate for most or all of the analyses, so an alternative correction was employed for these data (see below).

Results

Fifty-seven analyses were obtained from 55 zircons, with two grains (4 and 11) each analysed twice. Results are listed in Table 1, and shown in a concordia diagram (Fig. 4) and a probability density diagram (Fig. 5).

Interpretation

Most analyses are concordant or only slightly discordant, prior to correction for common Pb (Fig. 4). In view of the inapplicability of the common-Pb correction method that uses measured ^{204}Pb (Figs 2 and 3), the 207-correction (which assumes precise concordance of the true radiogenic $^{238}U/^{206}Pb^*$ and $^{207}Pb^*/^{206}Pb^*$ dates for each analysis) was employed, and the resulting $^{238}U/^{206}Pb^*$ data (Table 1) are considered to provide the best estimates of the ages of these zircons. Note that because most analyses are essentially concordant, the $^{238}U/^{206}Pb^*$ dates are not very sensitive to the choice of common-Pb correction method, nor do they differ significantly from the uncorrected $^{238}U/^{206}Pb$ dates.

Two analyses are characterized by high common-Pb contents ($f207$ >1%) based on the 207-correction. The dates obtained from these two analyses (Group D; Table 1) are imprecise or unreliable, and are not considered geologically significant. The remaining 55 analyses can be divided into two groups, based on their 207-corrected $^{238}U/^{206}Pb^*$ ratios.

Group 1 comprises three analyses of three zircons (Table 1), which yielded a weighted mean $^{238}U/^{206}Pb^*$ date of 694 ± 13 Ma (MSWD = 0.57).

Group 2 comprises 52 analyses of 50 zircons (Table 1), which yielded $^{238}U/^{206}Pb^*$ dates of 1606–820 Ma.

It is possible that all analyses in Group 1 represent unmodified detrital zircons, in which case the $^{238}U/^{206}Pb^*$ date of 689 ± 11 Ma (1σ) for analysis 45.1 represents a maximum age for deposition of the sandstone. However, the date of 694 ± 13 Ma for the three analyses in Group 1 represents a more robust estimate for the maximum depositional age.

The 52 analyses in Group 2 indicate $^{238}U/^{206}Pb^*$ dates that define significant age components (based on three or more data points) at c. 827, c. 869, and c. 1176 Ma, and a minor component at 1597 Ma (Fig. 5). These are interpreted as the ages of zircon-bearing rocks in the detrital source region(s) of the quartz sandstone.

Table 1. Ion microprobe analytical results for zircons from sample 181127: quartz sandstone, Durba Hills

Grp no.	Spot no.	Grain .spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f207 (%)	$^{238}U/^{206}Pb$ ±1σ	$^{207}Pb/^{206}Pb$ ±1σ	$^{238}U/^{206}Pb^*$ ±1σ	$^{207}Pb^*/^{206}Pb^*$ ±1σ	$^{238}U/^{206}Pb^*$ date (Ma) ±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ	Disc (%)
1	47	45.1	152	78	0.53	0.244	8.850 ± 0.148	0.06443 ± 0.00132	8.872 ± 0.152	– –	689 ± 11	– –	–
1	38	36.1	161	77	0.49	0.355	8.826 ± 0.147	0.06540 ± 0.00134	8.858 ± 0.152	– –	690 ± 11	– –	–
1	46	44.1	124	127	1.06	0.372	8.630 ± 0.150	0.06592 ± 0.00146	8.663 ± 0.155	– –	704 ± 12	– –	–
2	20	4.2	238	145	0.63	0.105	7.369 ± 0.139	0.06728 ± 0.00079	7.376 ± 0.144	– –	820 ± 15	– –	–
2	11	11.1	110	69	0.65	0.200	7.355 ± 0.148	0.06810 ± 0.00178	7.370 ± 0.154	– –	820 ± 16	– –	–
2	21	11.2	134	78	0.60	0.328	7.272 ± 0.146	0.06940 ± 0.00106	7.296 ± 0.151	– –	828 ± 16	– –	–
2	4	4.1	236	131	0.57	0.132	7.247 ± 0.137	0.06792 ± 0.00079	7.257 ± 0.141	– –	832 ± 15	– –	–
2	39	37.1	198	160	0.84	0.130	7.045 ± 0.113	0.06862 ± 0.00108	7.055 ± 0.117	– –	855 ± 13	– –	–
2	52	50.1	126	125	1.02	0.148	6.916 ± 0.119	0.06924 ± 0.00133	6.926 ± 0.123	– –	869 ± 14	– –	–
2	51	49.1	164	68	0.43	0.324	6.813 ± 0.112	0.07106 ± 0.00120	6.835 ± 0.117	– –	880 ± 14	– –	–
2	42	40.1	61	42	0.71	0.058	6.693 ± 0.135	0.06943 ± 0.00193	6.697 ± 0.141	– –	897 ± 18	– –	–
2	56	54.1	116	108	0.96	0.682	5.280 ± 0.093	0.08234 ± 0.00141	5.316 ± 0.098	– –	1111 ± 19	– –	–
2	13	13.1	132	167	1.31	0.720	5.166 ± 0.101	0.08357 ± 0.00103	5.204 ± 0.108	– –	1133 ± 21	– –	–
2	35	33.1	139	160	1.19	0.628	5.106 ± 0.087	0.08330 ± 0.00131	5.139 ± 0.093	– –	1146 ± 19	– –	–
2	57	55.1	261	239	0.95	0.097	5.101 ± 0.079	0.07905 ± 0.00091	5.106 ± 0.083	– –	1153 ± 17	– –	–
2	14	14.1	86	78	0.94	0.639	5.057 ± 0.102	0.08382 ± 0.00121	5.089 ± 0.109	– –	1157 ± 23	– –	–
2	25	23.1	190	129	0.70	0.266	5.071 ± 0.082	0.08067 ± 0.00106	5.084 ± 0.086	– –	1158 ± 18	– –	–
2	40	38.1	59	60	1.05	0.661	5.051 ± 0.101	0.08405 ± 0.00187	5.084 ± 0.108	– –	1158 ± 22	– –	–
2	18	18.1	194	147	0.78	0.336	5.066 ± 0.097	0.08128 ± 0.00084	5.083 ± 0.102	– –	1158 ± 21	– –	–
2	24	22.1	117	154	1.36	0.726	5.043 ± 0.088	0.08461 ± 0.00138	5.080 ± 0.093	– –	1158 ± 20	– –	–
2	55	53.1	65	63	1.00	0.728	5.038 ± 0.100	0.08459 ± 0.00188	5.075 ± 0.107	– –	1159 ± 22	– –	–
2	31	29.1	224	121	0.56	0.360	5.032 ± 0.078	0.08176 ± 0.00091	5.050 ± 0.083	– –	1165 ± 18	– –	–
2	22	20.1	179	205	1.19	0.462	5.022 ± 0.082	0.08267 ± 0.00110	5.046 ± 0.087	– –	1166 ± 18	– –	–
2	7	7.1	133	120	0.94	0.248	5.025 ± 0.098	0.08089 ± 0.00156	5.038 ± 0.104	– –	1167 ± 22	– –	–
2	33	31.1	184	183	1.03	0.155	5.027 ± 0.082	0.08014 ± 0.00110	5.035 ± 0.086	– –	1168 ± 18	– –	–
2	23	21.1	103	151	1.52	0.474	5.004 ± 0.090	0.08289 ± 0.00146	5.028 ± 0.095	– –	1169 ± 20	– –	–
2	41	39.1	359	285	0.82	0.329	4.999 ± 0.075	0.08180 ± 0.00076	5.015 ± 0.079	– –	1172 ± 17	– –	–
2	27	25.1	96	94	1.00	0.372	4.982 ± 0.090	0.08226 ± 0.00150	5.001 ± 0.096	– –	1175 ± 21	– –	–
2	2	2.1	88	96	1.13	–0.042	4.997 ± 0.101	0.07882 ± 0.00111	4.995 ± 0.106	– –	1177 ± 23	– –	–
2	10	10.1	128	145	1.17	0.361	4.975 ± 0.099	0.08223 ± 0.00096	4.993 ± 0.105	– –	1177 ± 23	– –	–
2	28	26.1	256	411	1.66	0.163	4.980 ± 0.085	0.08061 ± 0.00092	4.988 ± 0.090	– –	1178 ± 19	– –	–
2	50	48.1	220	190	0.89	0.124	4.977 ± 0.090	0.08032 ± 0.00123	4.983 ± 0.095	– –	1179 ± 21	– –	–
2	54	52.1	104	106	1.05	0.224	4.970 ± 0.087	0.08118 ± 0.00138	4.981 ± 0.093	– –	1179 ± 20	– –	–
2	34	32.1	65	55	0.87	0.426	4.959 ± 0.099	0.08294 ± 0.00184	4.980 ± 0.105	– –	1180 ± 23	– –	–
2	17	17.1	131	135	1.06	0.080	4.968 ± 0.097	0.08004 ± 0.00095	4.972 ± 0.102	– –	1181 ± 22	– –	–
2	32	30.1	74	79	1.11	0.404	4.951 ± 0.096	0.08280 ± 0.00177	4.971 ± 0.102	– –	1182 ± 22	– –	–
2	19	19.1	86	70	0.85	0.230	4.956 ± 0.100	0.08135 ± 0.00116	4.967 ± 0.106	– –	1182 ± 23	– –	–
2	12	12.1	130	108	0.86	0.128	4.959 ± 0.097	0.08052 ± 0.00099	4.965 ± 0.103	– –	1183 ± 22	– –	–
2	53	51.1	84	86	1.06	0.306	4.948 ± 0.108	0.08200 ± 0.00156	4.963 ± 0.115	– –	1183 ± 25	– –	–
2	9	9.1	125	130	1.08	0.215	4.947 ± 0.097	0.08133 ± 0.00149	4.957 ± 0.102	– –	1185 ± 22	– –	–
2	8	8.1	102	126	1.28	0.303	4.936 ± 0.098	0.08213 ± 0.00111	4.951 ± 0.104	– –	1186 ± 23	– –	–
2	30	28.1	176	131	0.77	0.207	4.926 ± 0.080	0.08143 ± 0.00108	4.936 ± 0.085	– –	1189 ± 19	– –	–
2	37	35.1	179	139	0.80	0.009	4.935 ± 0.080	0.07976 ± 0.00106	4.936 ± 0.084	– –	1189 ± 19	– –	–

Table 1. (continued)

Grp no.	*Spot no.*	*Grain .spot*	^{238}U *(ppm)*	^{232}Th *(ppm)*	$^{232}Th/^{238}U$	*f207 (%)*	$^{238}U/^{206}Pb$ ±1σ	$^{207}Pb/^{206}Pb$ ±1σ	$^{238}U/^{206}Pb^*$ ±1σ	$^{207}Pb^*/^{206}Pb^*$ ±1σ	$^{238}U/^{206}Pb^*$ *date (Ma)* ±1σ	$^{207}Pb^*/^{206}Pb^*$ *date (Ma)* ±1σ	*Disc (%)*
2	6	6.1	162	129	0.82	–0.018	4.924 ± 0.095	0.07965 ± 0.00086	4.923 ± 0.100	– –	1192 ± 22	– –	–
2	29	27.1	106	84	0.82	0.274	4.899 ± 0.088	0.08220 ± 0.00146	4.912 ± 0.093	– –	1194 ± 21	– –	–
2	48	46.1	209	136	0.67	0.048	4.905 ± 0.080	0.08034 ± 0.00098	4.907 ± 0.084	– –	1196 ± 19	– –	–
2	36	34.1	93	125	1.39	–0.167	4.914 ± 0.089	0.07857 ± 0.00149	4.906 ± 0.095	– –	1196 ± 21	– –	–
2	15	15.1	165	190	1.19	0.099	4.901 ± 0.094	0.08079 ± 0.00093	4.906 ± 0.099	– –	1196 ± 22	– –	–
2	3	3.1	82	68	0.86	–0.039	4.888 ± 0.102	0.07980 ± 0.00117	4.886 ± 0.108	– –	1200 ± 24	– –	–
2	43	41.1	89	77	0.89	0.226	4.864 ± 0.108	0.08212 ± 0.00151	4.875 ± 0.114	– –	1203 ± 26	– –	–
2	44	42.1	65	59	0.94	0.220	4.850 ± 0.130	0.08220 ± 0.00175	4.861 ± 0.138	– –	1206 ± 31	– –	–
2	1	1.1	96	90	0.98	0.204	4.835 ± 0.096	0.08223 ± 0.00191	4.845 ± 0.103	– –	1210 ± 23	– –	–
2	49	47.1	265	90	0.35	0.162	4.623 ± 0.071	0.08400 ± 0.00083	4.631 ± 0.075	– –	1260 ± 19	– –	–
2	45	43.1	34	29	0.88	0.106	3.594 ± 0.081	0.09862 ± 0.00218	3.598 ± 0.090	– –	1581 ± 35	– –	–
2	5	5.1	111	94	0.87	0.142	3.529 ± 0.069	0.10028 ± 0.00107	3.534 ± 0.076	– –	1606 ± 30	– –	–
D	16	16.1	63	79	1.30	1.063	9.325 ± 0.243	0.06995 ± 0.00290	9.425 ± 0.253	– –	650 ± 17	– –	–
D	26	24.1	224	310	1.43	1.855	5.234 ± 0.083	0.09234 ± 0.00108	5.333 ± 0.089	– –	1108 ± 17	– –	–

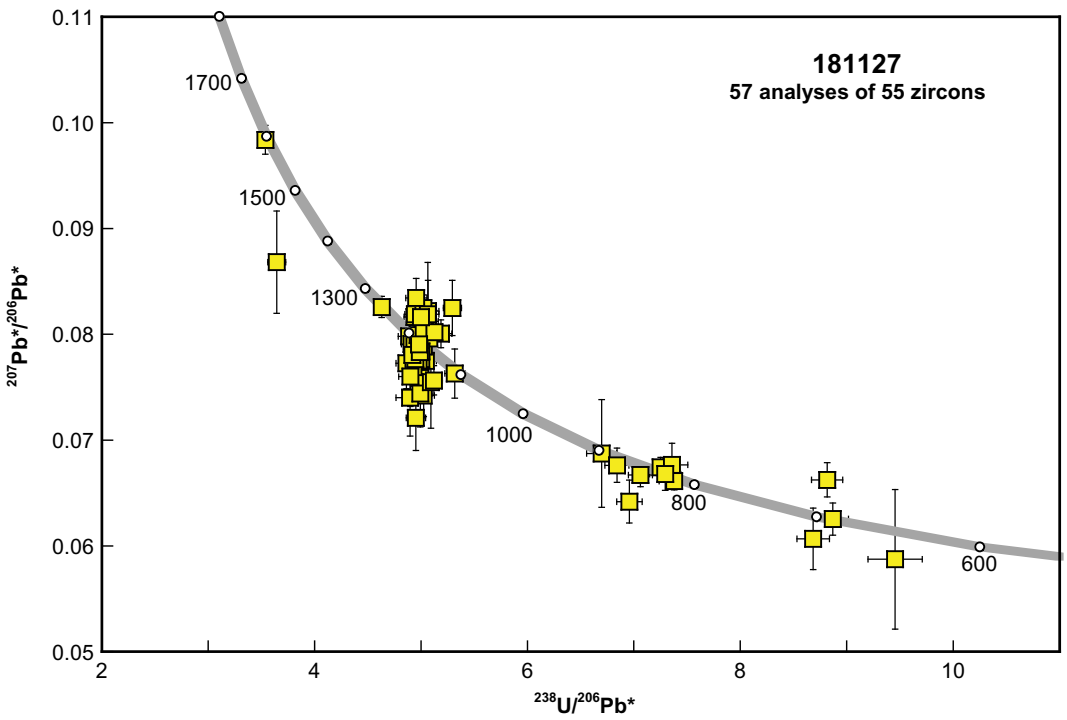

Figure 2. U–Pb analytical data for sample 181127: quartz sandstone, Durba Hills. All data are corrected for common Pb using measured ^{204}Pb, and exhibit significant dispersion in ^{207}Pb*/^{206}Pb*

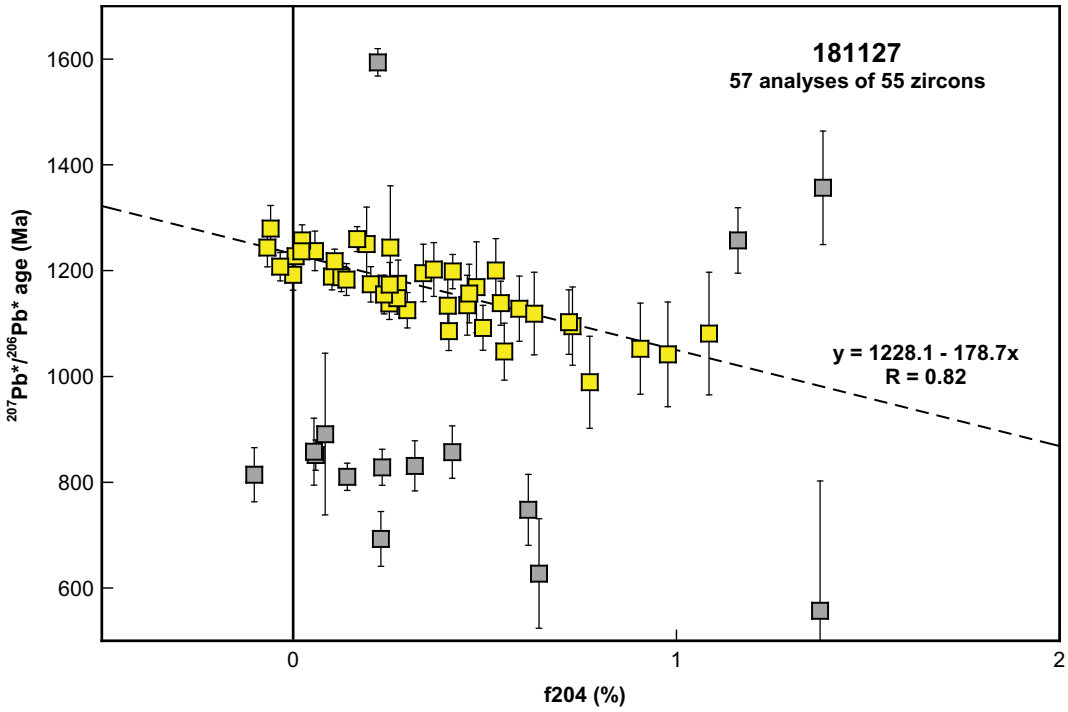

Figure 3. Correlation between ƒ204 and ^{207}Pb*/^{206}Pb* age (corrected for common Pb using measured ^{204}Pb) for sample 181127: quartz sandstone, Durba Hills. Yellow squares indicate data included in the linear regression, and the equation of the best-fit line is shown. R is Pearson's correlation coefficient

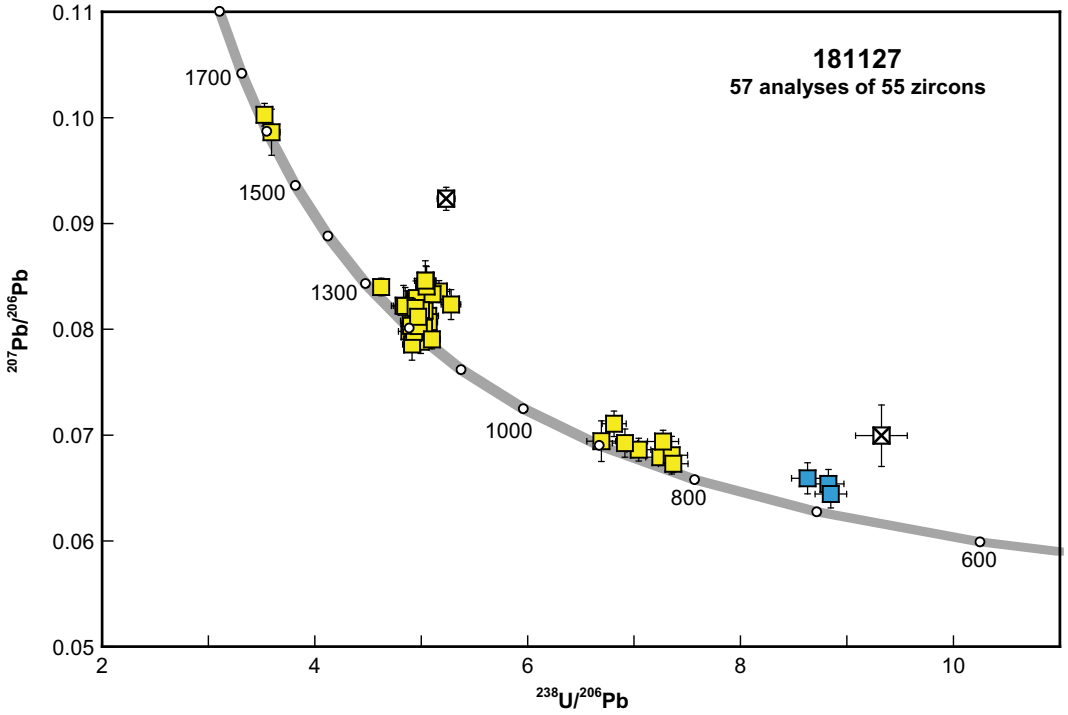

Figure 4. U–Pb analytical data for sample 181127: quartz sandstone, Durba Hills. Data are not corrected for common Pb. Blue squares indicate Group 1 (youngest detrital zircons); yellow squares indicate Group 2 (older detrital zircons); crossed squares indicate ungrouped analyses (*f*207 >1%)

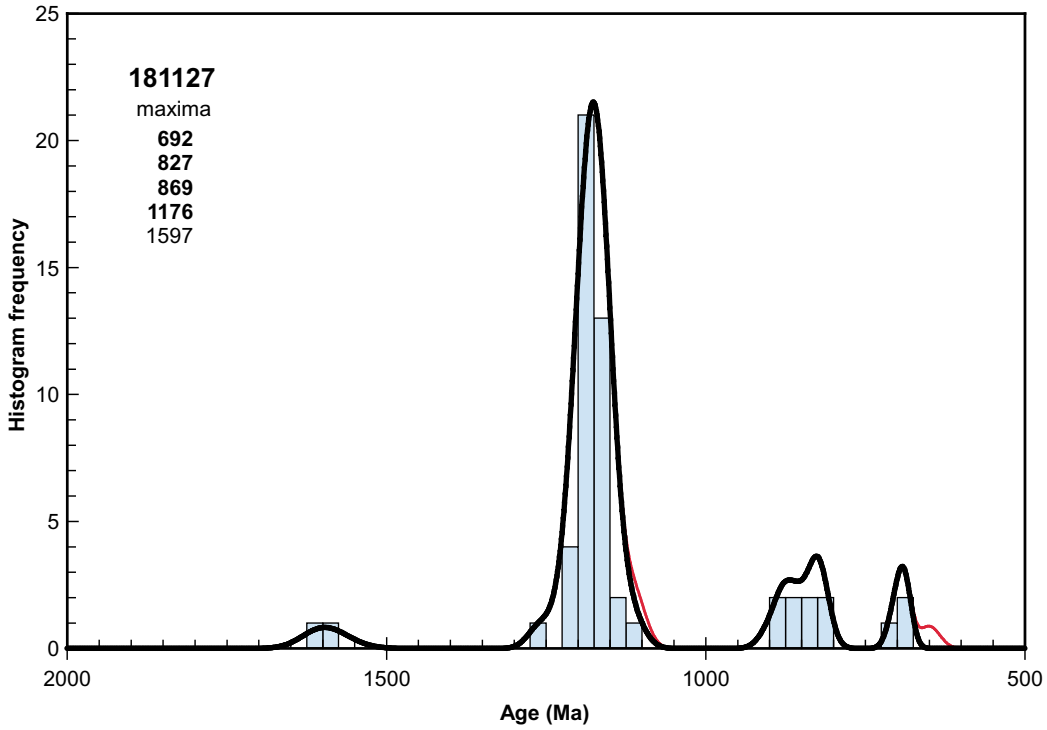

Figure 5. Probability density diagram and histogram of 207-corrected $^{238}U/^{206}Pb^*$ data from sample 181127: quartz sandstone, Durba Hills. Heavy curve, maxima values, and frequency histogram (bin width 25 Ma) include only data with *f*207 <1% (55 analyses of 53 zircons). Light curve includes all data (57 analyses of 55 zircons)

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